

Jeffery Liu

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Third-year BE(Hons) Mechatronics Engineering student at the University of Canterbury, graduating late 2027. NZ permanent resident with full working rights. Hands-on mechanical design and fabrication background, from an R&D internship at Holmes Solutions to rebuilding engines and running my own licensed vehicle dealership. Seeking a 2026/27 internship in hands-on mechanical design, tooling and process engineering.

KEY SKILLS

Technical

- **Mechanical Design & Verification: SolidWorks CAD and FEA**, hand calculations, GD&T, rapid prototyping, construction drawings. Designed a **50+ tonne rated retrofit** for an existing test apparatus at Holmes Solutions and flagged a **non-conservative capacity check** for structural review.
- **Workshop & Fabrication: MIG and arc welding, machining**, plasma and oxy cutting, **precision engine assembly**. Built through the Holmes internship, an engine rebuild, and repairing 35+ cars in my own workshop.
- **Electrical Design & Test: PCB design** (KiCad), SMT assembly, soldering, **wiring harness installation and connector repair**, electrical diagnosis. Applied on the UC Motorsport tyre-temperature sensor, university robotics projects, engine swaps and marine wiring.

Professional

- **Ownership & Management**: Ran a **licensed vehicle dealership solo** for three years alongside full-time study, covering sourcing, pricing, compliance and customers. **Sourced and scoped contractors** for the Holmes build.
- **Communication**: Researched and designed to validate **OCIMF MEG-4** guidelines, **justified designs to senior engineers**, and co-authored a **conference paper**. Consult directly with tuning clients and deal with buyers and sellers as a car dealer. Bilingual in **English and Mandarin**.

EXPERIENCE

Mechatronics Engineering Intern, Holmes Solutions

Nov 2025 – Feb 2026

Roadside & Marine sector, R&D consultancy (453.5 hours)

- Investigated mooring rope failures; designed a **sleeve retrofit** rated above **50 tonnes**, verified by **FEA**.
- Produced **construction drawings** and a **staged build plan**, then **sourced contractors** and material suppliers.
- Design was built and passed full-scale testing; co-authored the resulting **conference paper**.
- Assisted **dynamic testing**; practised **welding, machining** and **plasma cutting**.

Reference: Ray Keane, Senior Engineer, ray.keane@holmessolutions.com

Founder & Licensed Vehicle Dealer, Jeff's Auto Retail

Apr 2023 – Jun 2026

- Registered as a sole trader at 18 as a **self-funded** student.
- Ran the business from my own workshop instead of a yard, prioritising **fast turnover**.
- Sourced, diagnosed, repaired and resold **35+ cars** solo, to WOF standard.

Junior Vehicle Dynamics Engineer, UC Motorsport (FSAE)

Apr 2026 – present

Junior lead on the tyre-temperature sensing system project

- Selected **IR pyrometry**; owned schematic, **PCB layout** (KiCad) and components for an **ESP32** module.
- Assembled the board via **SMT** pick-and-place and reflow, wrote **CircuitPython** firmware, and designed a housing **prototype** built to handle 2.5+ G; first track test produced clean data.

Performance Consultant & Tuning Support, Custom Culture & Tuning Jan 2025 – present

- Consult clients on parts and performance; support an Auckland tuner with **datalogging** and adjustments.

Reference: Amit Lal, Tuner/Business Owner, cct.tune@gmail.com

Bicycle Technician & Maintenance Tutor, Adventure Cycles Apr 2019 – Oct 2023

- Diagnosed, serviced and repaired all bicycle components, and taught public **maintenance workshops**.

Bike Mechanic, Evo Cycles / Pitcrew Cycles Oct 2021 – Jan 2023

- Assembled bikes and carried out mechanical servicing during school holidays.

Workshop Apprentice, Steelguard Ltd Mar 2023 – Aug 2023

- Assisted during a post-flood relocation, gaining machine shop and welding experience.

PROJECTS

Turbocharged Engine Rebuild & Swap Dec 2025 – May 2026

- Diagnosed an internal engine failure and rebuilt with **stronger internals** and bolt-ons to hit target power.
- Hand-assembled **precision-machined components**, verifying free movement and no excess play.
- Validated through **datalogging** and tuner collaboration; now daily driven.

RoboCup Autonomous Robot, Team University Project May 2026 – present

- Proposed the concept and lead **mechanical design** in a team of three; built the intake and sorting mechanism.
- Laminated **PETG sheet** onto rubber intake flaps, stiffening them without adding bulk or losing grip.
- Now focused on **autonomous navigation** and reliability; also developing electrical and software.

Line Follower Robot, Team University Project Aug 2025 – Oct 2025

- Led **rapid-prototyped** mechanical design, powered by a custom PCB with motor driver and PD control.
- 3D-printed chassis and drivetrain with 22:1 **two-stage gearbox** and **silicone tyre production**.
- Set a practice lap fast enough to win; finished 3rd after a late motor failure.

Fixed Wing UAV Nov 2025 – Jan 2026

- Designed a **reflexed airfoil** with root-to-tip washout to **improve stall behaviour** on a flying-wing layout.
- Selected the propulsion and electronics; built a **hot-wire cutter** to cut the foam wing cores to **exact profile**.

Pick and Place CNC Machine, Team University Project Jul 2025 – Aug 2025

- Team of five: Prototyped and built a sub-\$200 pick-and-place CNC with a **waterjet-cut** aluminium frame.
- Delivered for \$128.85, assembled in 1 hour, positioning within 0.3 mm.

EDUCATION

University of Canterbury, BE(Hons) Mechatronics Engineering Feb 2024 – present

- Third year, expected graduation late 2027. Offered two competitive UC entrance scholarships.

ADDITIONAL INFORMATION

Work Rights: New Zealand Permanent Resident, full right to work in New Zealand with no restrictions.

Availability: Full-time from mid-November 2026.

Qualifications: NZ Workplace First Aid Certificate, Unit standards 6400, 6401, 6402, March 2026.

Hobbies: cars, boating, fishing, camping, snowboarding, cooking.